

Siddharth G

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EDUCATION

National Institute of Technology, Tiruchirappalli
B.Tech. in Electronics and Communication Engineering

Tiruchirappalli, TN
Nov 2022 – Present

EXPERIENCE

Software Engineering Intern

May 2025 – Jul 2025

Oracle

Bangalore, Karnataka

- Developed **Java Spring Boot REST APIs** for a Gen2 Bulk Cleanup Tool, automating cleanup across **1,000+ OCI tenancies** and replacing manual Selenium workflows.
- Achieved **99.5% execution accuracy** by building secure, token-authenticated microservices that identify and remove unused cloud resources.
- Built usage-fetching APIs for analytics and billing pipelines and authored **JUnit test suites**, ensuring fail-safe deployments.

ML Research Intern

May 2024 – Mar 2025

AGS Health

Tiruchirappalli, TN

- Architected an end-to-end **ML-driven pipeline** automating structural analysis and conversion of **200+** complex documents into editable files.
- Engineered a **DBSCAN-based** pre-processing module for column detection, header/footer extraction, and table recognition using **Python, OpenCV, and scikit-learn**.
- Improved **Document Layout Analysis accuracy by 60%** using multi-modal layout analysis and high-fidelity OCR/table-recognition models.

PROJECTS

High-Performance URL Shortener | *TypeScript, Hono, PostgreSQL, Drizzle ORM, Redis, BullMQ* Jul 2025

- Built a URL-shortening service using **Base62 encoding** and **PostgreSQL** for collision-free short URL generation.
- Implemented a **Redis cache-aside layer** for redirect lookups, reducing direct database reads and improving redirect latency.
- Designed an async analytics pipeline using **BullMQ** and Redis queues to process click events without blocking redirects.
- Added **Redis-backed rate limiting**, click tracking, and visitor analytics APIs; deployed frontend on **Vercel**, backend on **Render**, with **Neon PostgreSQL** and **Upstash Redis**.

FinanceX | *Next.js, TailwindCSS, OAuth, Hugging Face*

Mar 2025

- Engineered a full-stack personal finance assistant with secure multi-user **OAuth** and **JWT** authentication.
- Automated receipt and PDF ingestion with **Tesseract.js**, **pdf-parse**, and **Hugging Face** models to extract transaction data automatically.
- Built interactive dashboards with **Recharts** to visualize spending patterns and financial insights.
- Implemented a **modular, scalable architecture**, ensuring robust performance across web and mobile platforms.

MERNBot | *TypeScript, MongoDB, Node.js, Express.js, React.js, JWT*

Feb 2024

- Built a full-stack AI chatbot with a **React** frontend and **Node.js/Express** backend integrating the GPT-3.5 API.
- Implemented secure authentication using **bcrypt** password hashing and **JWT**-based session management.
- Developed a responsive, **WebSocket**-enabled frontend with **Vite.js** for real-time bidirectional chat.
- Designed **RESTful API** endpoints connecting frontend and backend services for reliable data flow.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL, HTML5/CSS

Backend & Frameworks: Spring Boot, Node.js, Express.js, Hono, MERN, JUnit

Data & ML: OpenCV, SciPy, scikit-learn, PyTesseract

Databases & Infra: PostgreSQL, MongoDB, Redis, Drizzle ORM, Oracle Cloud (OCI)

Tools: Git, Postman, Figma, Jira, Google Colab

Certifications: IBM ML with Python, Google AI Essentials, AI/ML on Google Cloud, UX Design Foundations